Chapter 7

Wired and Wireless Local Area Networks

Chapter 7: Outline

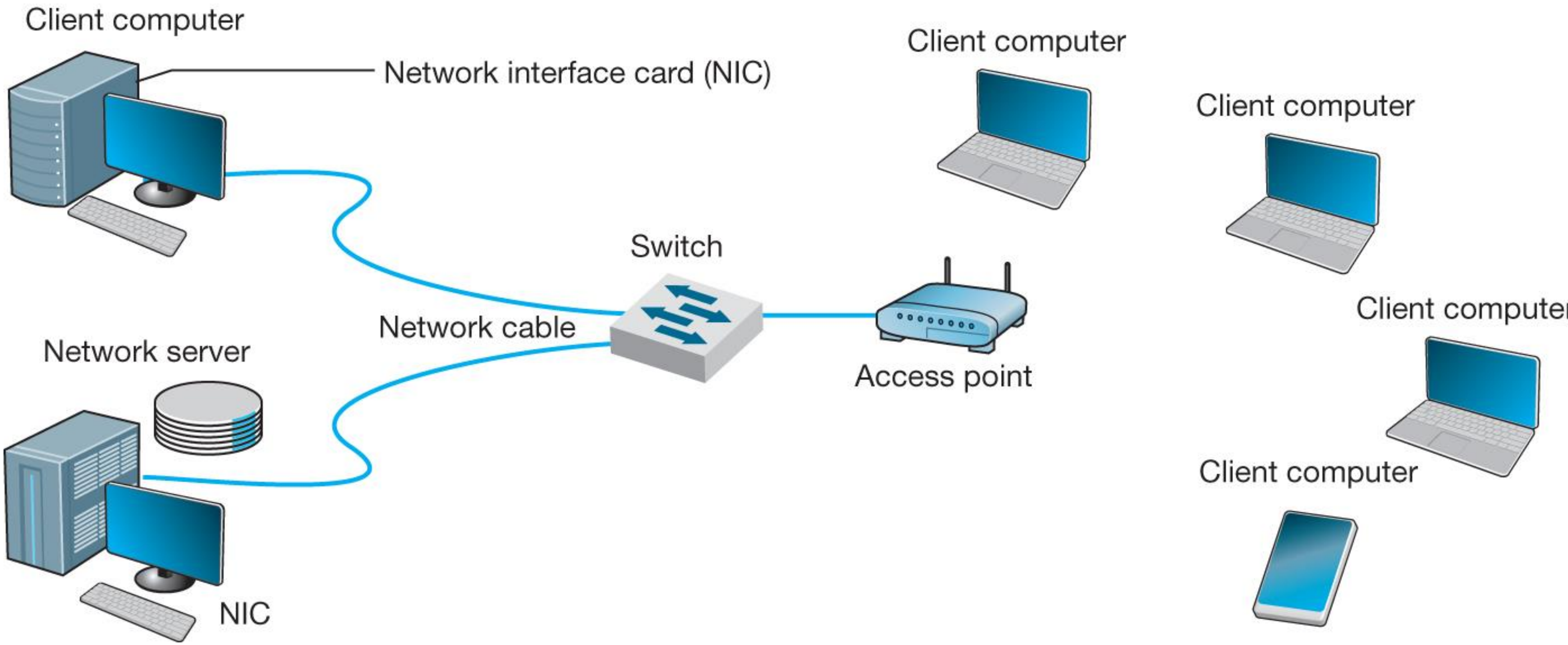
Why Use a LAN?

- Information sharing
  - Having users access the same files, exchange information via email, or use Internet
    - Ex: single purchase order database accessed by all users on the LAN
  - Results in improved decision making
- Resource sharing
  - Having hardware devices shared by all users
    - Printers, Internet connections
  - Having software packages shared by all users on a LAN
  - Results in reduced cost

Sharing Software on a LAN

- Purchase software on a per seat basis
  - Install software on a server for all to use
  - No need to have a copy on every computer on the LAN
  - Reduces cost
  - Simplifies maintenance and upgrades
  - Example
    - LAN: a 30 client network
    - Purchase only a 10-seat license for a software program (instead of purchasing 30 copies of the same program)
    - Assumes that only 10 users would simultaneously use the software

7.2 LAN Components



Network Interface Cards (NICs)

- **Also called network cards and network adapters**
- **Contains physical and data link layer protocols**
  - **Includes a unique data link layer address (called a MAC address), placed in them by their manufacturer**
  - **Includes a socket allowing computers to be connected to the network**
  - **Organizes data into frames and then sends them out on the network**
- **Mostly built into motherboards today**
- **Can be plugged into the USB port**

Hubs & Switches

Functions:

1) Act as junction boxes, linking cables from several computers on a network
    1) Usually sold with 4, 8, 16 or 24 ports
    2) May allow connection of more than one kind of cabling, such as UTP and coax.

2) Repeat (reconstruct and strengthen) incoming signals
    - Important since all signals become weaker with distance
    - Extends the maximum LAN segment distance

Hubs & Switches

a) Small-Office, Home-Office (SOHO) switch
with five 10/100/1000 ports

http://homestore.cisco.com/en-us/Switches/
linksys-EZXS55W_stcVVproductId53934575VVcatId543809VVviewpro

b) Data center chassis switch with 512 10 Gbps ports

Source: newsroom.cisco.com/dlls/2008/prod_012808b.html

Access Points

- **APs are used instead of hubs/switches in a wireless environment**
- **Act as a repeater**
  - **They must be able to hear all computers on a WLAN**

Access Points

(a) AP for SOHO use

(b) A Power-Over-Ethernet AP for enterprise use

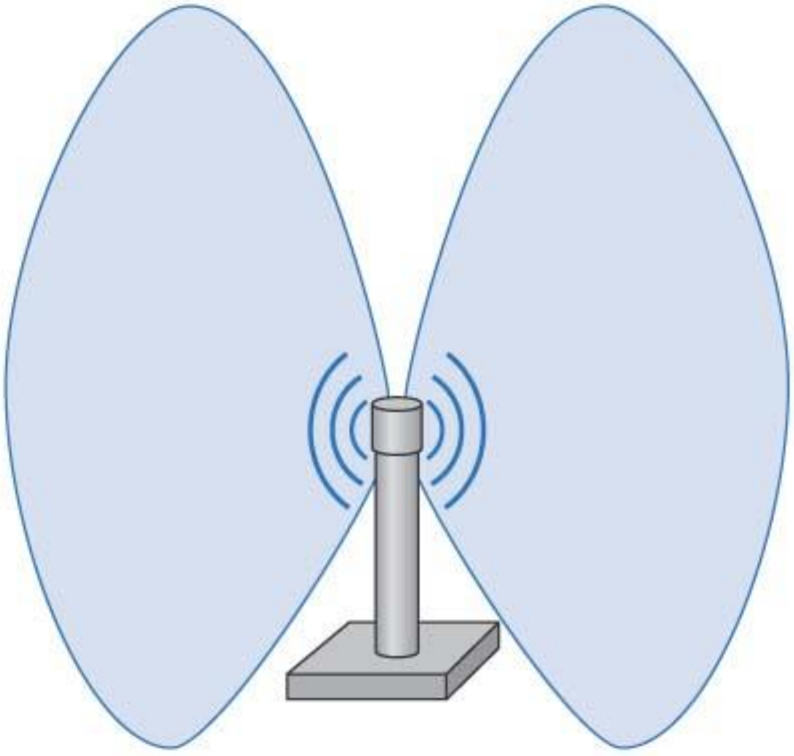
Access Points (Cont.)

- Power over Ethernet (POE)
  - Used to supply power to some APs
  - No external power is needed
  - Power flows over unused Cat5 wires

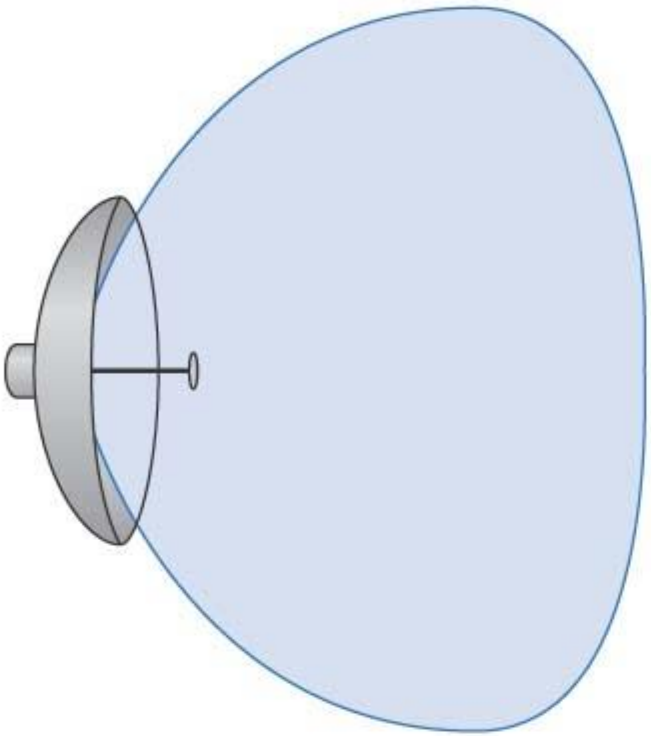
Antennas used in WLANs

- Omni directional antennas
  - Transmit in all directions simultaneously
  - Used on most WLANs
    - Dipole antenna
      - Transmits in all direction (vertical, horizontal, up, down)
- Directional antennas
  - Project signal only in one direction
    - Focused area; stronger signal; farther ranges
  - Most often used on inside of an exterior wall
    - To reduce the security issue
      - A potential problem with WLANs

Types of Antennas



(a) Omnidirectional antenna



(b) Directional antenna

7.3 Wired Ethernet

- Used by almost all LANs today
- Originally developed by a consortium of Digital Equipment Corp., Intel and Xerox
- Standardized as IEEE 802.3
- Types of Ethernet
  - Shared Ethernet
    - Uses hubs
  - Switched Ethernet
    - Uses switches

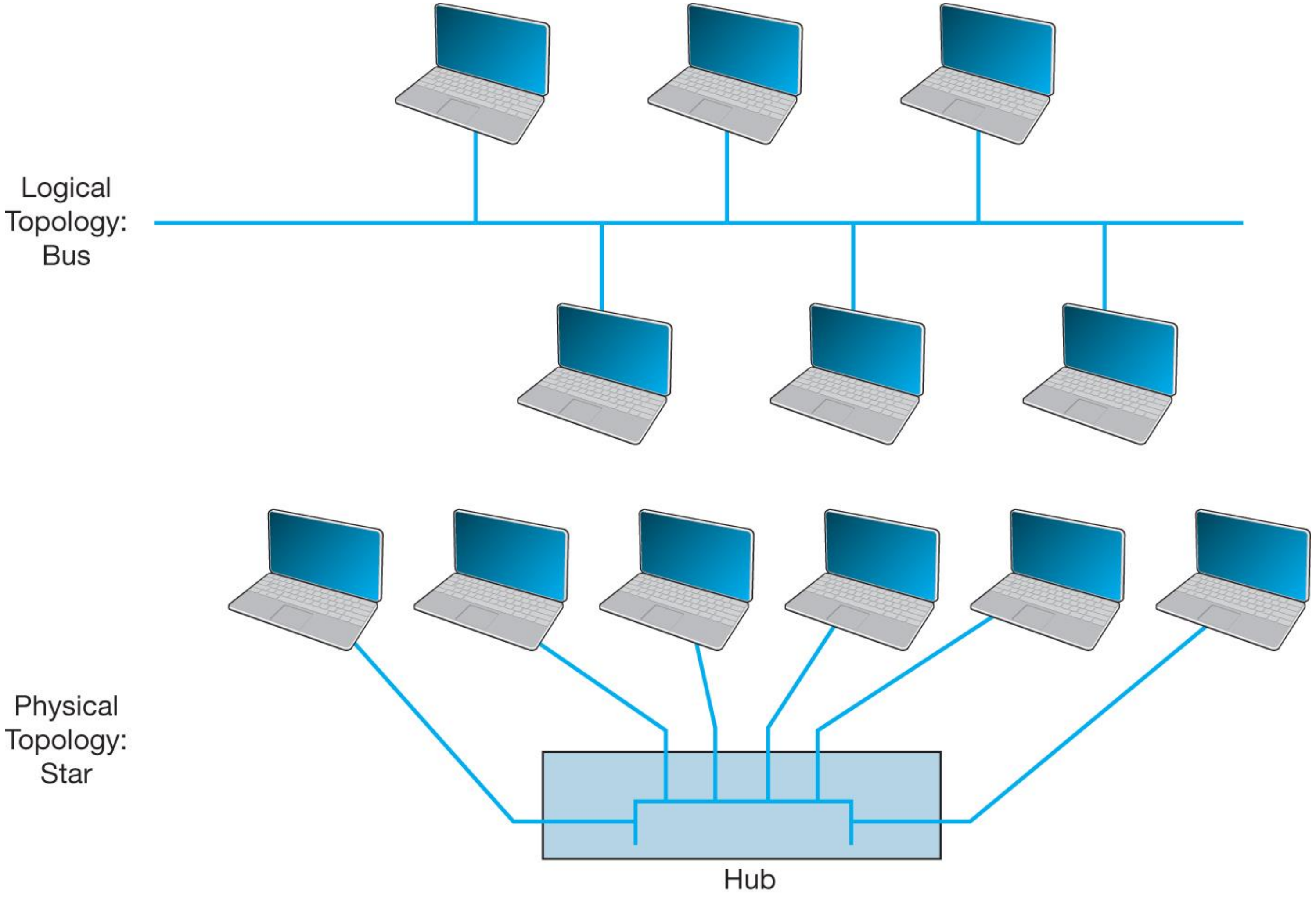
Topology

- Basic geometric layout of the network
  - The way computers on the network interconnected
- Logical Topology
  - How the network works conceptually
  - Like a logical data flow diagram (DFD) or
  - Like a logical entity relationship diagram (ERD)
- Physical Topology
  - How the network is physically installed
  - Like physical DFD or physical ERD

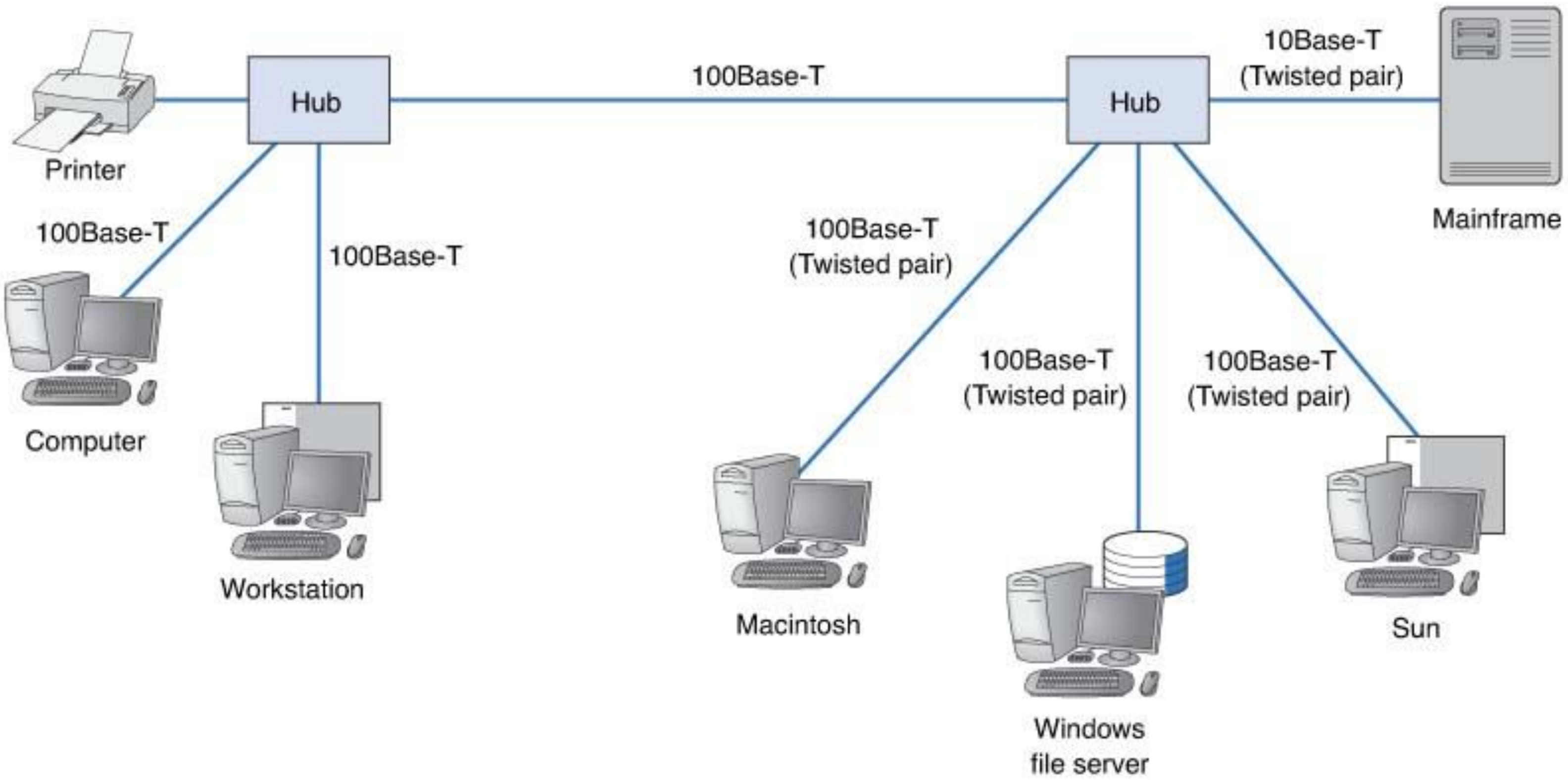
Shared Ethernet’s Logical Topology

- Viewed logically as a bus topology
- All messages from any computer flow onto the central cable (bus)
- A computer receive messages from all other computers, whether the message is intended for it or not
- When a frame is received by a computer, the first task is to read the frame’s destination address to see if the message is meant for it or not

Shared Ethernet's Physical Topology



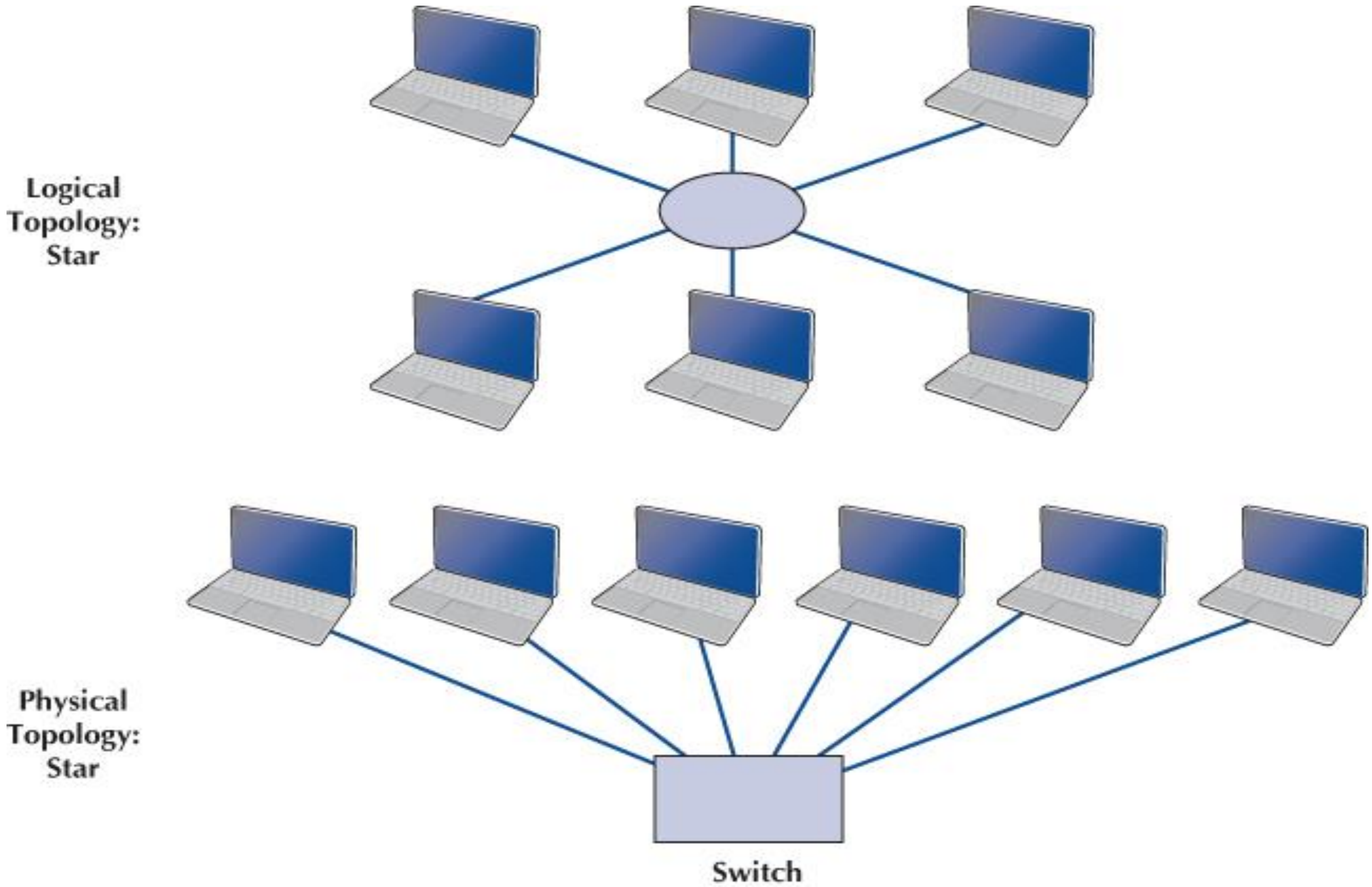
Multiple Hub Ethernet Design



Switched Ethernet Topology

- Uses workgroup switches
  - Designed to support a small set of computers (16 to 24) in one LAN
  - Looks similar to a hub, but very different inside
  - Designed to support a group of point-to-point circuits
    - No sharing of circuits
- Logical and physical topology of the network becomes a star topology via switch
- Switch reads destination address of the frame and only sends it to the corresponding port
  - While a hub broadcasts frames to all ports

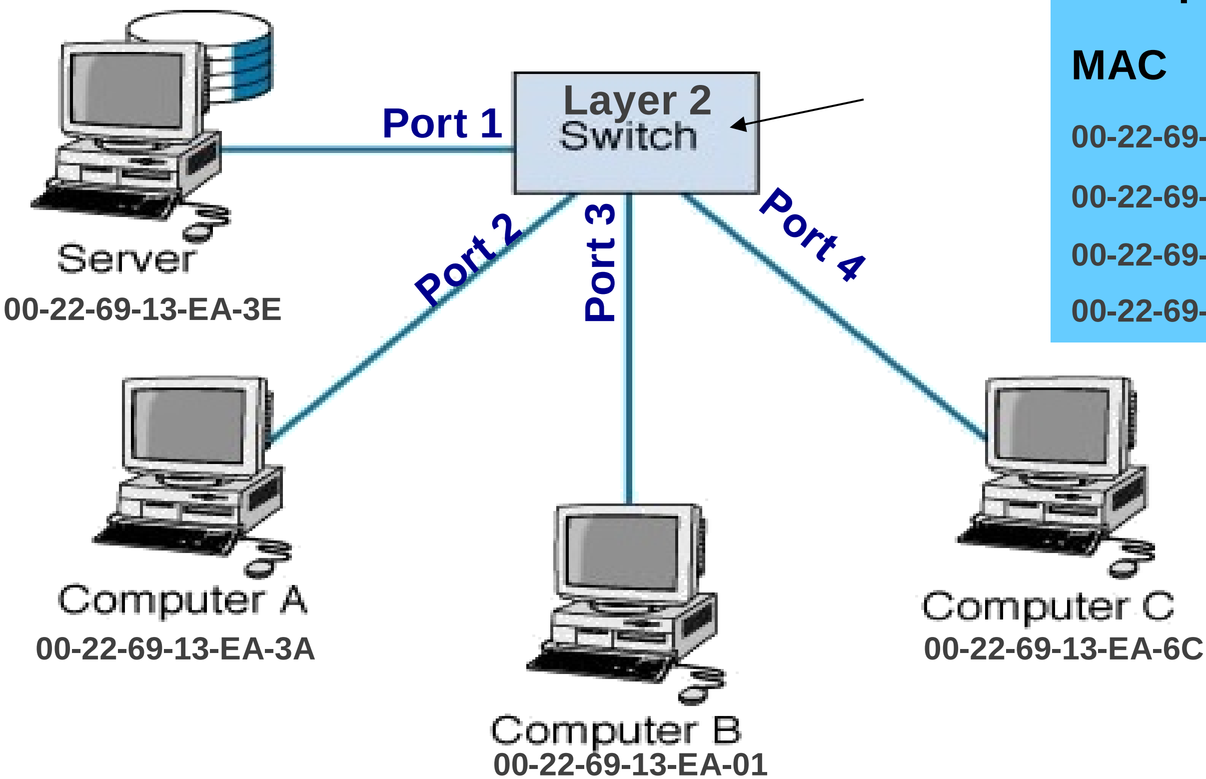
Switch Ethernet's Topologies



Forwarding Tables

- **Similar to routing tables**
- **Lists the Ethernet address of computers connected to each port**
- **When a frame is received, the switch reads its Layer 2 data link layer destination address and sends the frame out of the corresponding port in its forwarding table.**

Basic Switch Operation



Forwarding Table

MAC	Port
00-22-69-13-EA-3E	1
00-22-69-13-EA-3A	2
00-22-69-13-EA-01	3
00-22-69-13-EA-6C	4

Learning Switch Operation

- Switch starts by working like a simple hub
  - With an empty forwarding table
- It gradually fills its forwarding table by learning about the nodes
  - Reads the source MAC address of the incoming frame and records it to the corresponding port number
  - Reads the destination MAC address. If not in the Table then it broadcasts the frame to all ports
  - Waits for the destination computers to respond, and repeats the first step

Forwarding Table

MAC	Port
00-22-69-13-EA-3E	1
00-22-69-13-EA-3A	2
00-22-69-13-EA-01	3
00-22-69-13-EA-6C	4

Modes of Switch Operations

1. **Cut through switching**
   - Reads destination address and starts transmitting without waiting for the entire message to be received
   - Low latency; but may waste capacity (error messages)
   - Only on the same speed incoming and outgoing circuits
2. **Store and forward switching**
   - Waits until the entire frame is received, perform error control, and then transmit it
   - Less wasted capacity; slower network
   - Circuit speeds may be different
3. **Fragment free switching**
   - Reads the first 64 bytes (contains the header)
   - Performs error checking; if it is OK then begins transmitting
   - It is a compromise between previous two modes

Media Access Control (MAC) with Shared Ethernet

- Uses a contention-based protocol called CSMA/CD (Carrier Sense Multiple Access / Collision Detect)
- Frames can be sent by two computers on the same network at the same time
  - They will collide and destroy each other
  - Can be termed as “ordered chaos”
  - Tolerates, rather than avoids, collisions

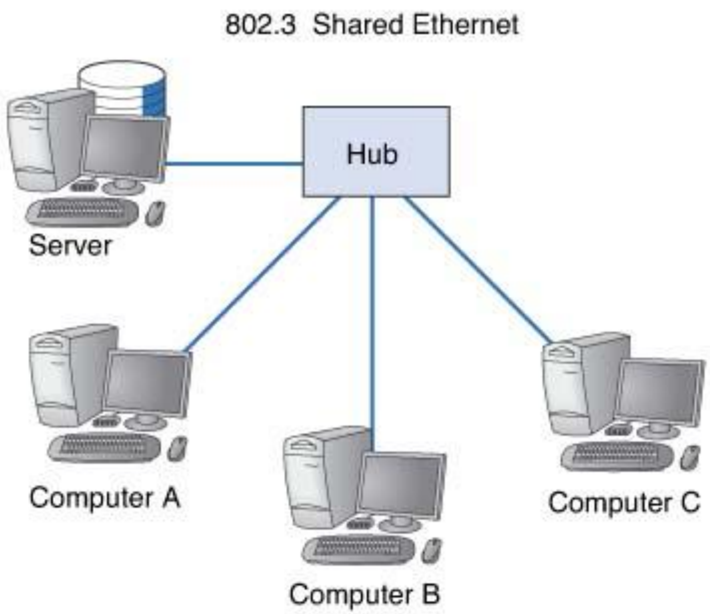
CSMA/CD

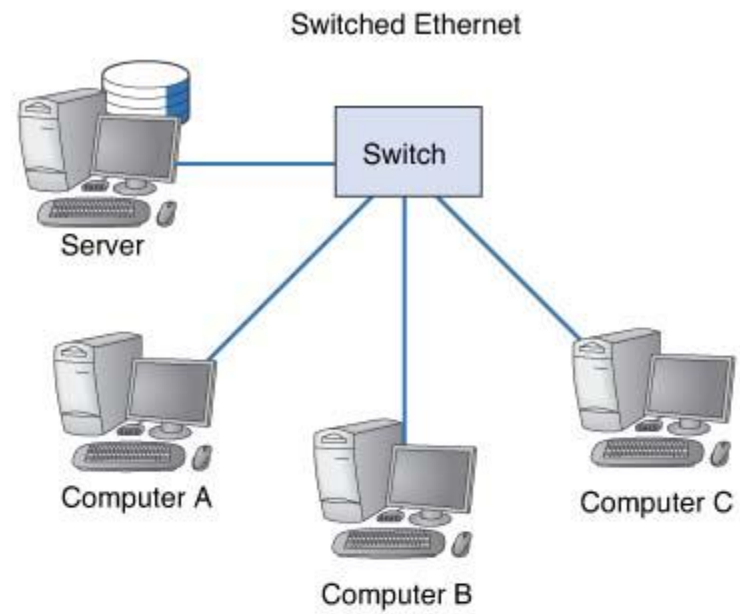
- **Carrier Sense (CS):**
  - A computer listens to the bus to determine if another computer is transmitting before sending anything
  - Transmit when no other computer is transmitting
- **Multiple Access (MA):**
  - All computers have access to the network medium
- **Collision Detect (CD):**
  - Declared when any signal other than its own detected
  - If a collision is detected
    - To avoid a collision, both wait a random amount of time and then resend message

Media Access Control (MAC) with Switched Ethernet

- Each circuit shared by a computer and the switch
- Still uses CSMA/CD media access control
  - Each device (computer or switch) listens before transmitting
- Multiple messages can be sent at the same time.
  - Computer A can send a message to computer B at the same time that computer C sends one to computer D
  - Two computers send frames to the same destination at the same time
  - Switch stores the second frame in memory until it finishes sending the first, then forwards the second

Performance Comparison





Capable of using about only 50% of capacity (10BaseT) before collisions become a problem

Runs at up to 95% capacity on 100Base-T

Twisted Pair Ethernets

- 10Base-T
  - Uses Cat 3 and Cat 5 UTP, very inexpensive
  - Runs up to 100 meters
  - Rapidly losing ground to 100Base-T
- 100Base-T
  - Uses Cat 5 UTP
  - Also called Fast Ethernet, replaced 10Base-T in sales volume
  - More common format today
- Combined 10/100 Ethernet
  - Some segments run 10Base-T and some run 100Base-T

Fiber Optic based Ethernets

- **1000Base-T (1 GbE)**
  - Gigabit Ethernet.
  - Maximum cable length is only 100 m for UTP cat5
  - Fiber Optic based (1000Base-LX) runs up to 440 meters
- **1000Base-F**
  - 1 Gbps fiber
- **10 GbE**
  - 10 Gbps Ethernet. Uses fiber and is typically full duplex
- **40 GbE**
  - 40 Gbps Ethernet. Uses fiber and is typically full duplex.

Summary - Ethernet Media Types

Name	Maximum Data Rate
10Base-T	10 Mbps
100Base-T	100 Mbps
1000Base-T	1 Gbps
1000Base-F	1 Gbps
10 GbE	10 Gbps
40 GbE	40 Gbps
100 GbE	100 Gbps

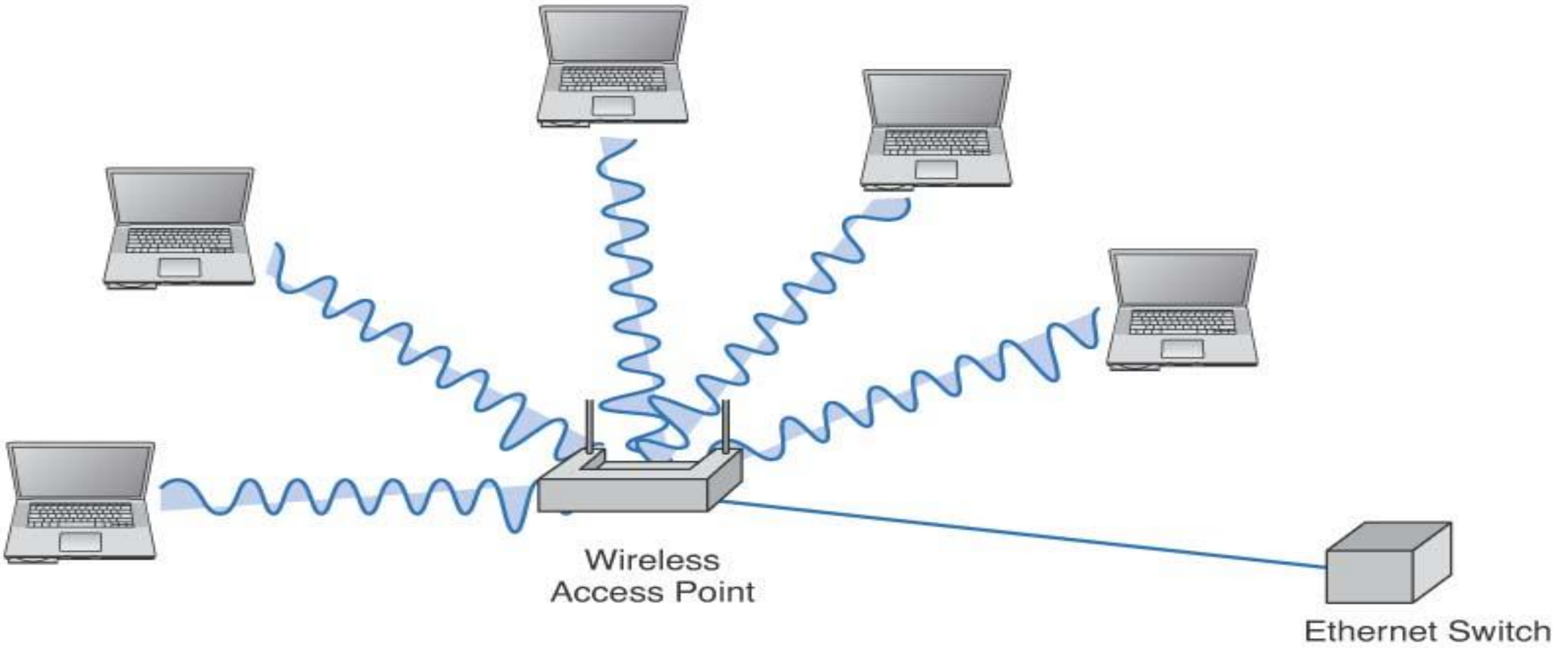
	IEEE	Gbits/sec	Fiber type	Number of fibers	Max link length (m)	Max channel insertion loss (in dB)
10-Gbit Ethernet	802.3ae	10	OM3	2	300	2.6
40-Gbit Ethernet	P802.3ba	40	OM3	8	100	1.9
40-Gbit Ethernet	P802.3ba	40	OM4	8	150	1.5
100-Gbit Ethernet	P803.2ba	100	OM3	20	100	1.9
100-Gbit Ethernet	P802.3ba	100	OM4	20	150	1.5

7.4 Wireless Ethernet

- Use radio frequencies to transmit signals through the air (instead of cables)
  - 802.1x family of standards (aka, Wi-Fi)
- Wi-Fi grown in popularity
  - Eliminates cabling
  - Facilitates network access from a variety of locations
  - Facilitates for mobile workers (as in a hospital)
  - Used in 90 percent of companies

WLAN Topology

Same as Ethernet
- **Physical star**
- **Logical bus**



WLAN Media Access Control

- **Uses CSMA/CA**
  - **CA → collision avoidance**
  - **A station waits until another station is finished transmitting plus an additional random period of time before sending anything**

Association with an AP

- Scanning- searching for available Aps
- Two types
  - Active
    - NIC transmits probe frame on all active channels
    - AP responds with info to associate with it
  - Passive
    - NIC listens on all channel for beacon frame
    - NIC can use info in beacon frame to associate with AP

IEEE 802.11x

- Wi-Fi is one of the fastest changing areas in networking.
- Last two or three versions are used while all others are obsolete but may still be in use.
- Backward compatible with a, b, and g
- Disadvantage: one laptop using a, b, or g slows down access by all other laptops (even when they are using n)

WLAN Characteristics

- Two frequency ranges
  - 2.4 GHz
  - 5 GHz
  - More frequency is newly added such as 6 GHz.
- Distance range of 35-150 meters
- Channels are used to reduce interference

IEEE 802.11n

- 802.11n is an obsolete version, but it is still used because it is cheap.
- 3 channels of 450 Mbps each with maximum range of 100 meters, in practice speed and distance are lower.
- 3 channels are numbered 1, 6, and 11.
- Dual-band AP combines all channels into one dual-band channel that provide 600 Mbps.

IEEE 802.11ac

- 802.11ac is an current version.
- Run in 2.4 and 5 GHz simultaneously.
- Some products may use only 2.4 GHz, few channels and/or lower speed.
- RTS/CTS is sent on a separated frequency, so it does not interfere with data transmission.
- Modulation is 256 QAM (8 bits/symbol) increased from 64 QAM (6 bits/symbol) of 802.11n.

IEEE 802.11ac

- 8 channels of 433 Mbps each with maximum range of 50 meters.
  - But the actual throughput under the efficiency of data link protocol is about 300 Mbps.
- As client gets further from AP, the speed drops.
  - At maximum range, the speed is about 90 Mbps per channel (60 Mbps throughput)
  - At maximum speed, the range is only within 20-30 meters.

IEEE 802.11ax (Wi-Fi 6)

- Wireless Ethernet with three frequencies, 2.4, 5, and 6 GHz.
- 9.6 Gbps across 8 channels
  - 1.2 Gbps per channel under perfect conditions.
- Entire data stream can be combined into one large channel.
- Modulation is 1024-QAM (10 bits/symbol)
- Range is 35 meters but likely 7-10 meters
  - Short range is good for large office building where Aps are installed close together to serve many users in the same area.

IEEE 802.11ax (Wi-Fi 6)

- New innovation of 802.11ax is the ability for AP to combine packets for different devices into the same physical layer transmission.
  - The same physical transmission can contain Ethernet packets destined to different computers.
- The second innovation is the ability to “color” packets.
  - AP can tell that an incoming message is actually from another AP whose transmission is overlap, not one of its client in its WLAN.
    - AP can ignore the incoming packet and save a bit of processing time.

Types of Wireless Ethernet

	Channels	Max Bandwidth	Max range	Frequency	
802.11a	8	54 Mbps	50 meters	5 GHz	Legacy
802.11b	3	11 Mbps	150 meters	2.4 GHz	Legacy
802.11g	3	54 Mbps	150 meters	2.4 GHz	Legacy
802.11n	3 (1, 6, 11)	300 Mbps -or- 450 Mbps	100 meters	2.4/5 GHz	Dual-band AP
802.11ac	8	433 Mbps – 6.9 Gbps	50 meters	2.4/5 GHz	
802.11ax	8	9.6 Gbps	35 meters	2.4/5/6 GHz	“color”

WLAN Security

- Especially important for wireless network
  - Anyone within the range can use the WLAN
- Finding a WLAN
  - Move around with WLAN equipped device and try to pick up the signal
  - Use special purpose software tools to learn about WLAN you discovered
    - Wardriving – this type reconnaissance
    - Warchalking – writing symbols on walls to indicate presence of an unsecure WLAN

Types of WLAN Security

- Service Set Identifier (SSID)
  - Required by all clients to include this in every packet
  - Included as plain text →Easy to break
- Wired Equivalent Privacy (WEP)
  - Requires that user to enter a key manually (to NIC and AP)
  - Communications encrypted using this key
  - Requires user to log in.
  - Short key (40-128 bits) → Easy to break by "brute force"
  - One time WEP keys created dynamically after login
  - Requires a login (with password) to a server
  - Once client log out, the WEP key will be discarded.

Types of WLAN Security, cont'd

- Wi-Fi Protected Access (WPA)
  - Key can be fixed in AP or assigned dynamically as users login.
  - longer key, changed for every packet.
  - Each time the frame is transmitted, the key is changed.
- 802.11i (WPA2)
  - EAP login used to get session key
  - uses AES encryption
- MAC address filtering
  - Allows computers to connect to AP only if their MAC address is entered in the "accepted" list.
  - However, packet sniffer (such as Wireshark) can discover a valid MAC address. The hacker can use a software to change the address to one of the accepted list.

7.5 The Best Practice LAN Design

- Recently costs have dropped while speeds have increased
- WI-FI rates approaching that of wired
- WI-FI cheaper b/c of no wires to install
  - $150-400 per wire in retrofitted building
  - $50-$100 per wire in new building
- Best practice today: Wired Ethernet for primary LAN and WI-FI as overlay

Ethernet Switches	Price (each)
Ethernet 100Base-T 8-port switch	30
Ethernet 100Base-T 16-port switch	70
Ethernet 100Base-T 24-port switch	80
Ethernet 100Base-T 48-port switch	130
Ethernet 10/100/1000Base-T 8-port switch	70
Ethernet 10/100/1000Base-T 16-port switch	130
Ethernet 10/100/1000Base-T 24-port switch	200
Ethernet 10/100/1000Base-T 48-port switch	300
Upgrade any switch to include POE	75
Cable (Including Installation)	**Price (per cable)**
UTP Cat 5e (1000Base-T or slower)	50
UTP Cat 6 (1000Base-T or slower)	60
STP Cat 5e (1000Base-T or slower)	60
Wireless Access	**Price (each)**
802.11 wireless access point	60
802.11 wireless access point with POE	120
802.11 managed wireless access point with POE	200
Wi-Fi controller with POE 8-ports	300
Wi-Fi controller with POE 24-ports	400
Wi-Fi controller with POE 48-ports	500

Designing User Access with Wired Ethernet

- Switched 100Base-T over Cat5e
  - Relatively low cost and fast
- Category 5e cables
  - Costs decreasing
  - Provides room for upgrades to 100Base-T or 1000Base-T
- Install network cable during the construction of building
  - Adding cable to an existing building can cost significantly more.
  - Under construction, most building have a separate cable plan as they have plans for electrical cables.

Designing User Access with Wireless Ethernet

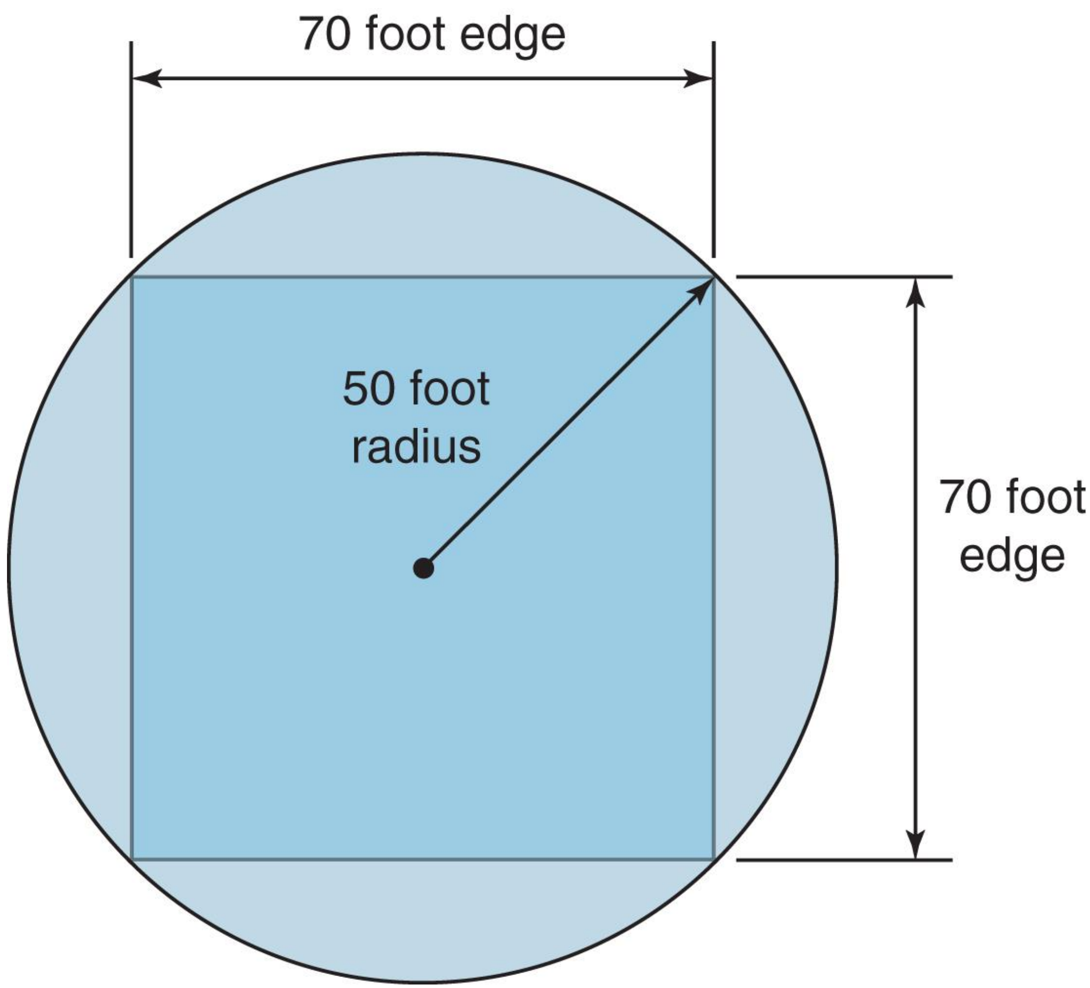
- Pick newest technology, cost permitting
  - 802.11ac or 802.11ax
- Placement of APs should be considered

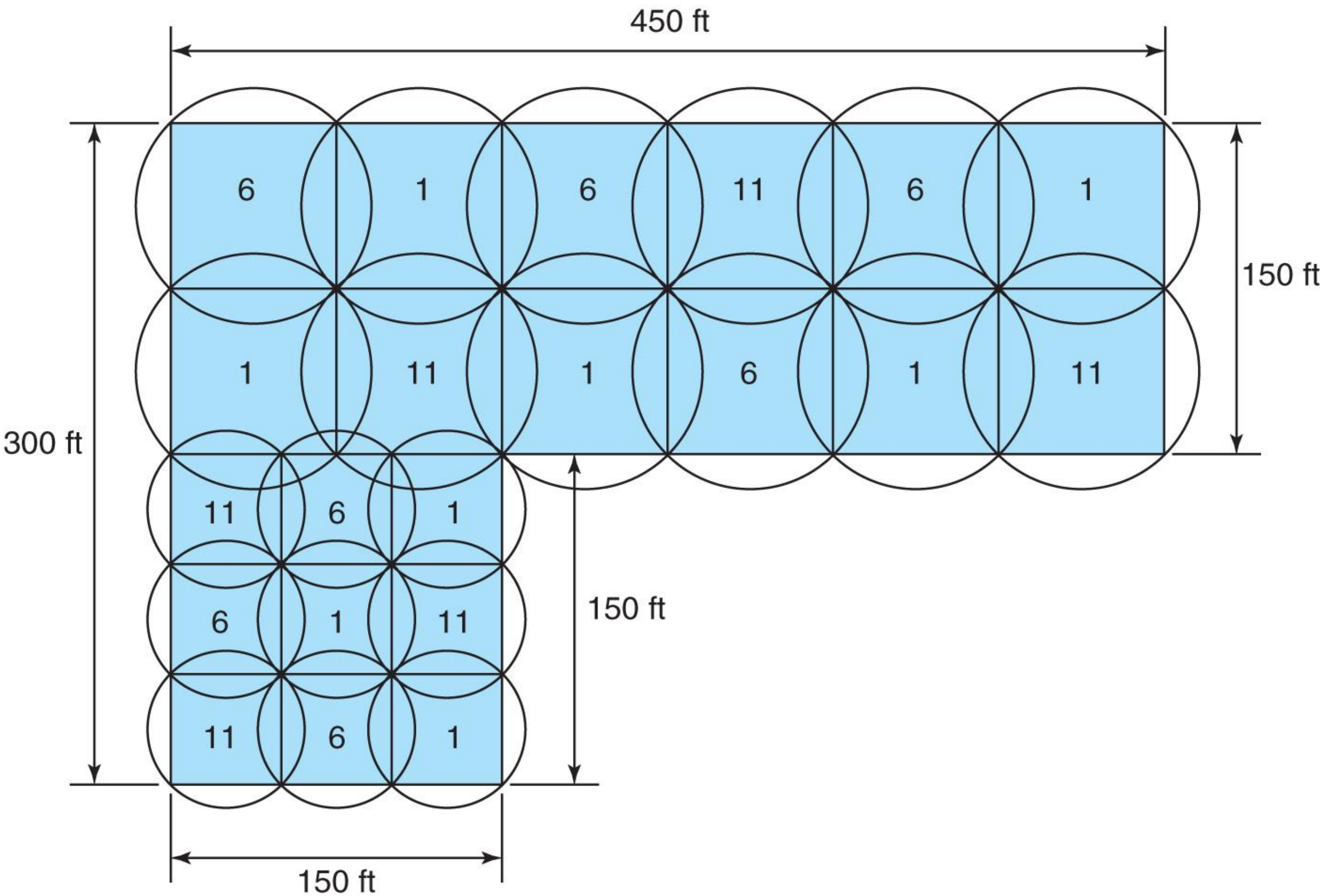
Physical WLAN Design

- More challenging than designing a traditional LAN
  - Use a temporary AP and laptop to evaluate placement of APs
  - Locations are chosen to provide coverage as well as to minimize potential interference
- Begin design with a site survey, used to determine:
  - Feasibility of desired coverage
    - Measuring the signal strength from temporary APs
  - Potential sources of interference
    - Most common source: Number and type of walls
  - Locations of wired LAN and power sources
  - Estimate of number of APs required

Physical WLAN Design

- Begin locating APs
  - Place an AP in one corner
  - Move around measuring the signal strength
  - Place another AP to the farthest point of coverage
    - AP may be moved around to find best possible spot
    - Also depends on environment and type of antenna
  - Repeat these steps several times until the corners are covered
  - Then begin the empty coverage areas in the middle
- Allow about 15% overlap in coverage between APs
  - To provide smooth and transparent roaming
- Set each AP to transmit on a different channel


70 foot edge
50 foot
radius
70 foot
edge


450 ft
6
1
6
11
6
1
1
11
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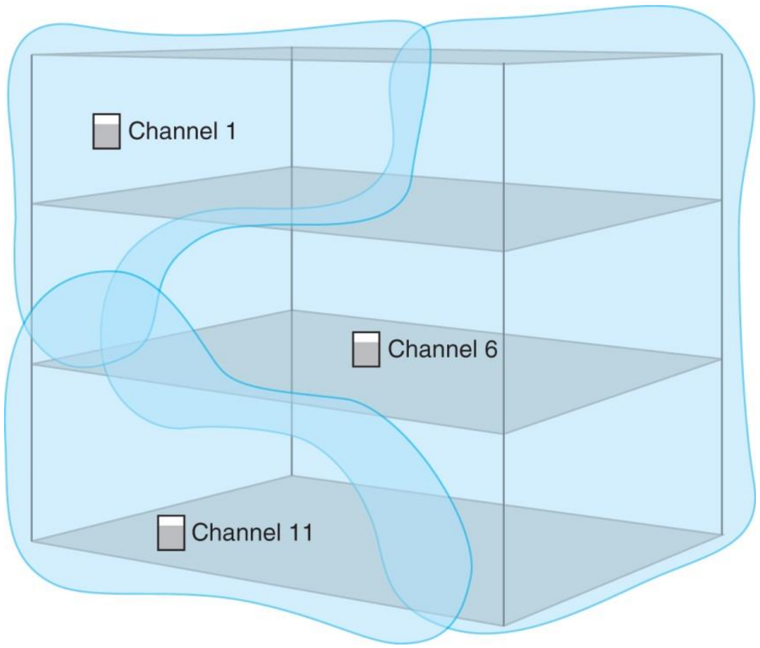
6
1
11
150 ft
300 ft
11
6
1
6
1
11
11
6
1
150 ft
150 ft

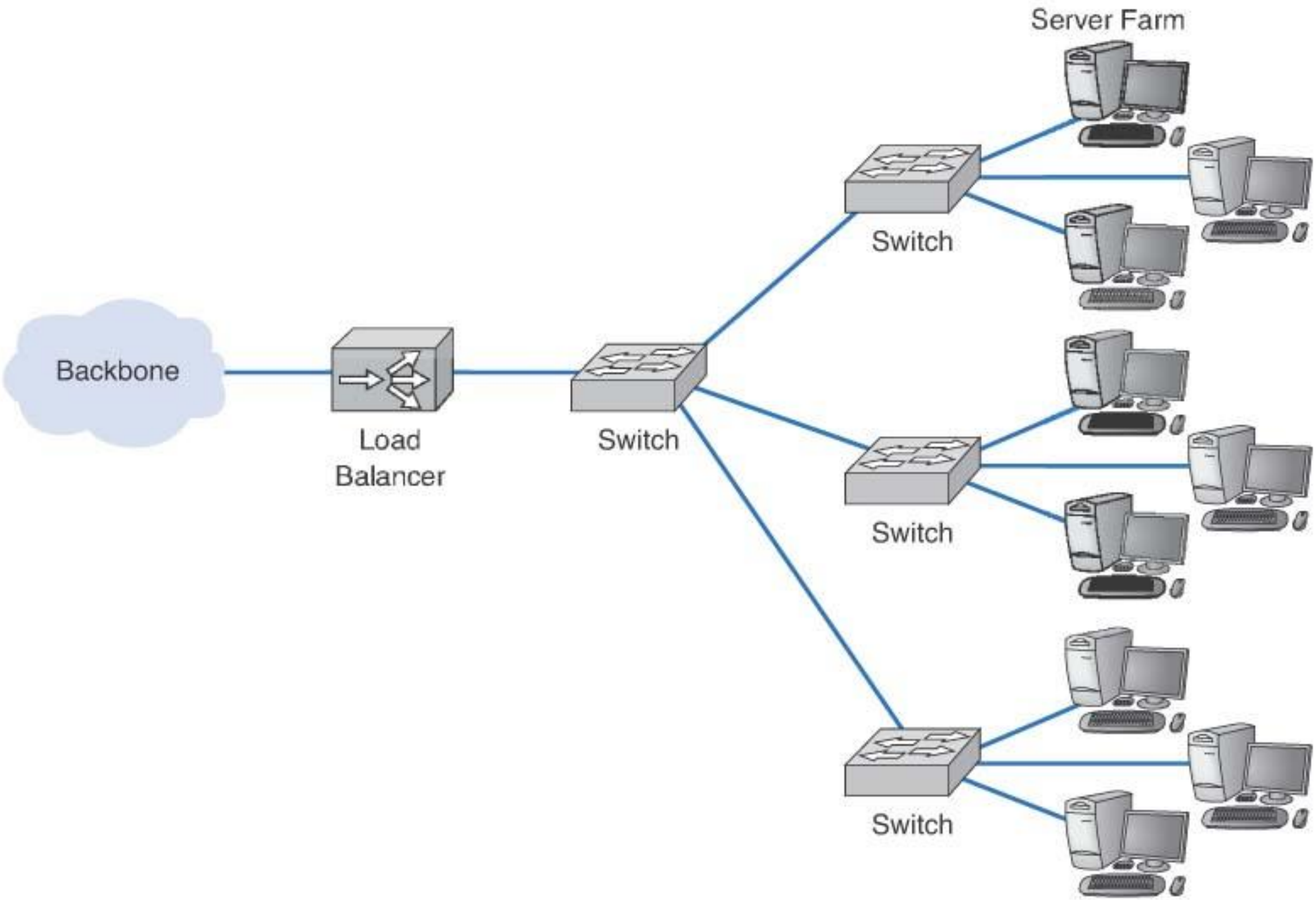
Multistory WLAN Design

- Must include
  - Usual horizontal mapping, and
  - Vertical mapping to minimize interference from APs on different floors



Designing Data Center

- Data center houses its primary server.
  - Servers are placed together in server farms or clusters.
  - Ensure that a request should be forward to a server that is not busy.
    - Load balancer or load balancing switch
      - Simple formula of round-robin
      - Complex formula to track how busy each server
- Server virtualization of creation of logically separate servers on the same physical server.


Server Farm
Backbone
Load
Balancer
Switch
Switch
Switch
Switch

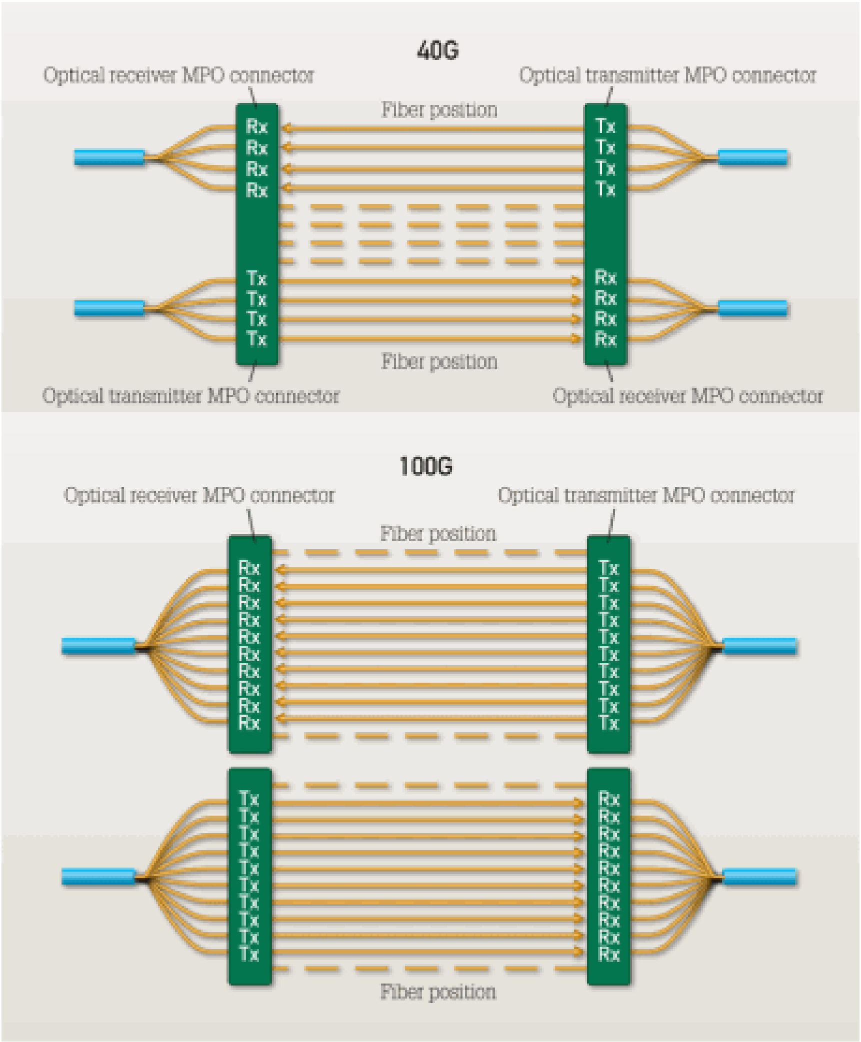
Data Flow in Data Center

- Storage Area Network, SAN, is a LAN devote only to data storage among

  High-speed storage <-> high-speed LAN <-> server

- Network-Attached Storage, NAS, devices
  - Not a general purpose computer
  - Large amount of disk storage with a small processor designed solely to response to requests for files or data
  - Attached to LAN where they function as fast file server

Fiber Optics Cable for Data Center



- 40-Gbit/sec parallel-optics transmission calls for 4x10G channels on four fibers per direction, which can be accommodated in a single 12-fiber connector.

- 100-Gbit/sec parallel-optics transmission calls for 10x10G channels on 10 fibers per direction.

http://www.cablinginstall.com/articles/print/volume-18/issue-11/features/data-center-migrating-to-40-and-100g-with-om3-and-om4-connectivity.html

7.6 Improving LAN Performance

- **Throughput:**
  - Used often as a measure of LAN performance
  - Total amount of user data transmitted in a given period of time
- **To improve throughput and LAN performance, identify and eliminate bottlenecks**
  - Bottlenecks are points in the network where congestion is occurring
  - Congestion is when the network or device can't handle all of the demand it is experiencing

Identifying Network Bottlenecks

- Potential places are server vs. circuit
  - Network server
  - Network circuit (especially LAN-BN connection)
  - Client's computer (highly unlikely, unless too old)
- How to find it
  - Check the server utilization during poor performance
    - If high >60%, then the server is the bottleneck
    - If low <40%, then the network circuit is the bottleneck
    - If between 40% - 60%, both the server and circuits are the bottlenecks

Improving Server Performance

- Software improvements
  - Choose a faster NOS
  - Fine tune network and NOS parameters such as
    - Amount of memory used for disk cache
    - Number of simultaneously open files
    - Amount of buffer space
- Hardware improvements
  - Add a second server
  - Upgrade the server’s CPU
  - Increase its memory space
  - Add more hard disks
  - Add a second NIC to the server

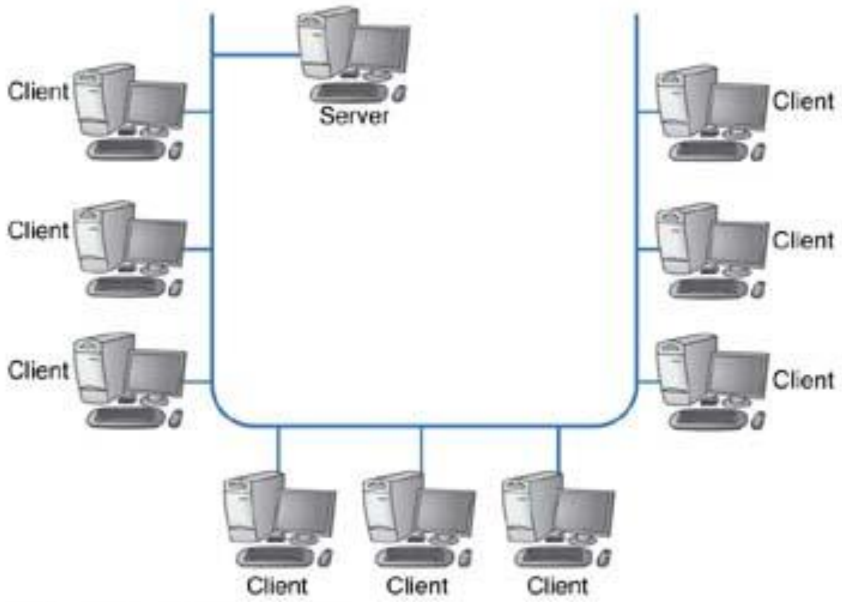
Improving Disk Drive Performance

- Especially important, since disk reads are the slowest task the server needs to do
- Consider Redundant Array of Inexpensive Disks (RAID)
  - Replacing one large drive with many small ones
  - Can be used to both improve performance and increase reliability
  - Building redundancy into the hard drives so drive failure does not result in any loss of data

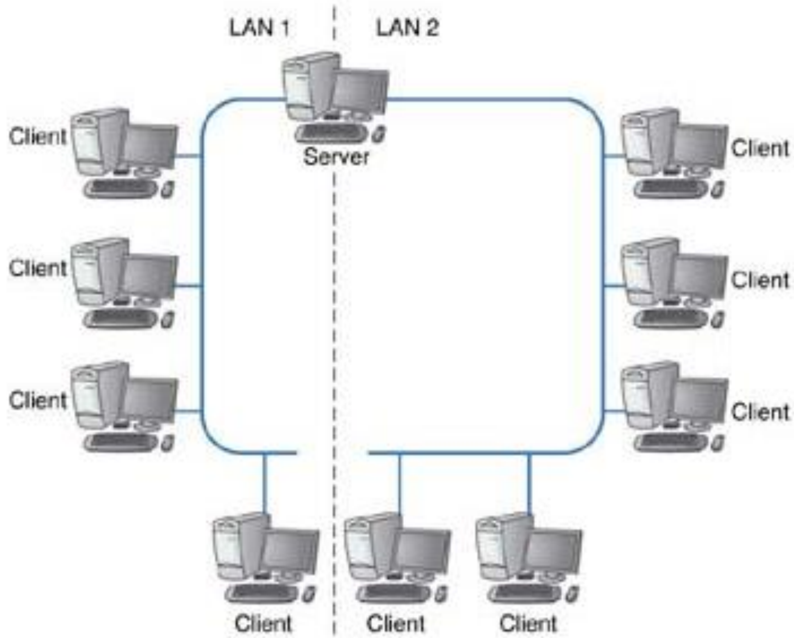
Improving Circuit Capacity

- Upgrade to a faster protocol
  - Means upgrading the NICs and possible cables
  - Examples:
    - Upgrading the network from 100Base-T to 1000Base-T
    - Upgrading the segment to the server from 100Base-T to 1000Base-T
- Increase number of circuits
  - Network segmentation
    - Wired
      - Add NICs to server that connect to multiple switches
    - Wireless
      - Add more APs on different channels

Network Segmentation



a Before network segmentation



b After segmentation

Reducing Network Demand

- Move files to client computers
  - Such as heavily used software packages
- Encourage balancing of wired and wireless network usage by users
- Move user demands to off peak times
  - Encourage users to not use the network as heavily during peak usage times such as early morning or after lunch
  - Delay some network intensive jobs to off-peak times, such as run heavy printing jobs at night

7.7 Implications for Cyber Security

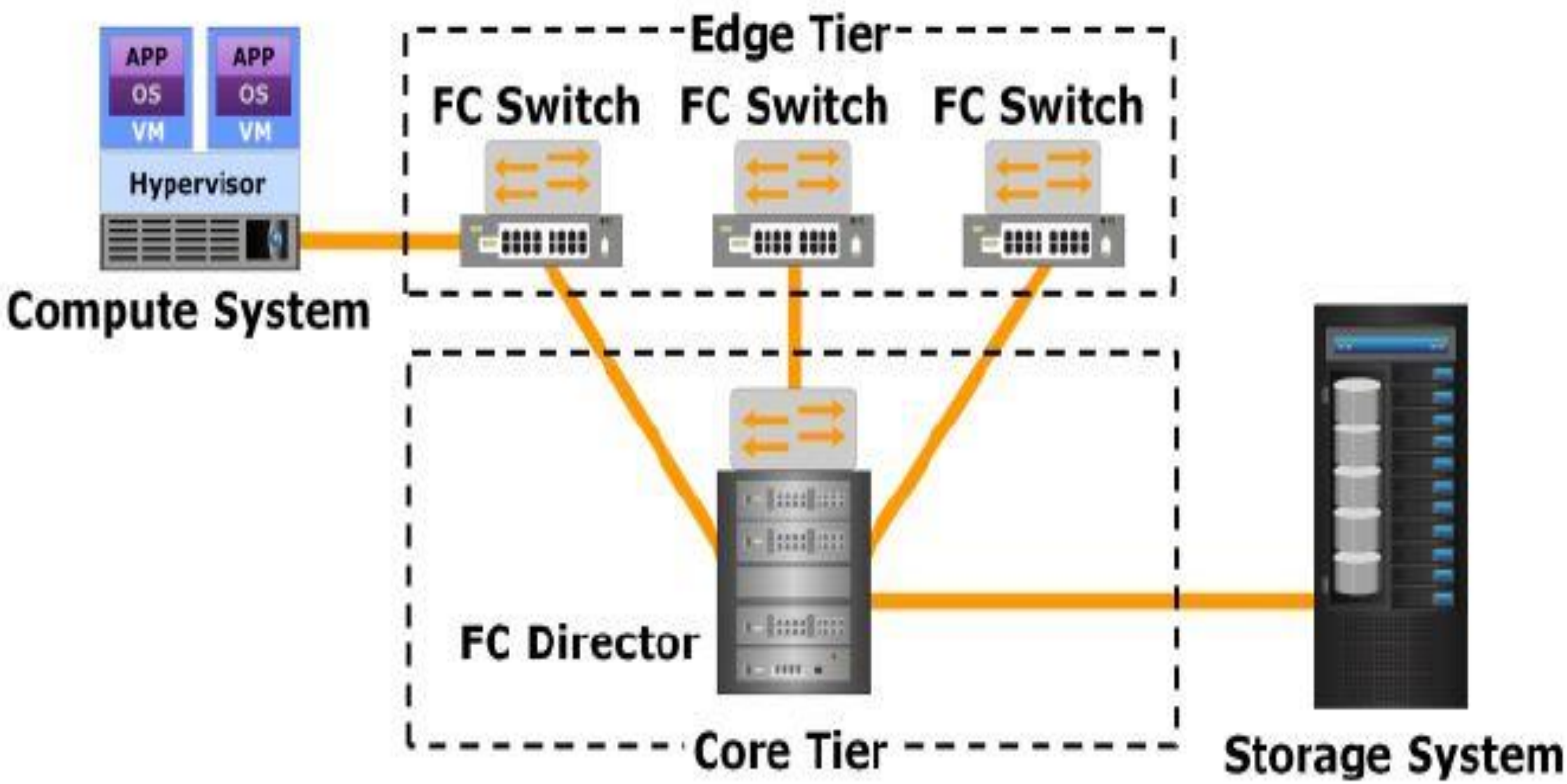
- **Secure LAN from unauthorized users.**
  - **All devices should be placed in locked closets.**
- **WLAN means to use encryption and user login.**
  - **Wi-Fi APs signal extends outside the building onto public street or other floors of occupied by other firms.**
  - **MAC address filter would allow only the register devices.**
- **Protect their internal network through network segmentation**
  - **Use different SSID (name of the network) to segment the WLAN**
  - **Some SSID for guest to offer Internet access with lower capacity (to deter people staying for too long.)**

Fibre Channel Switch

- Fibre Channel switch is a network switch compatible with the Fibre Channel (FC) protocol.
- It allows the creation of a Fibre Channel fabric, that is the core component of a storage area network (SAN).

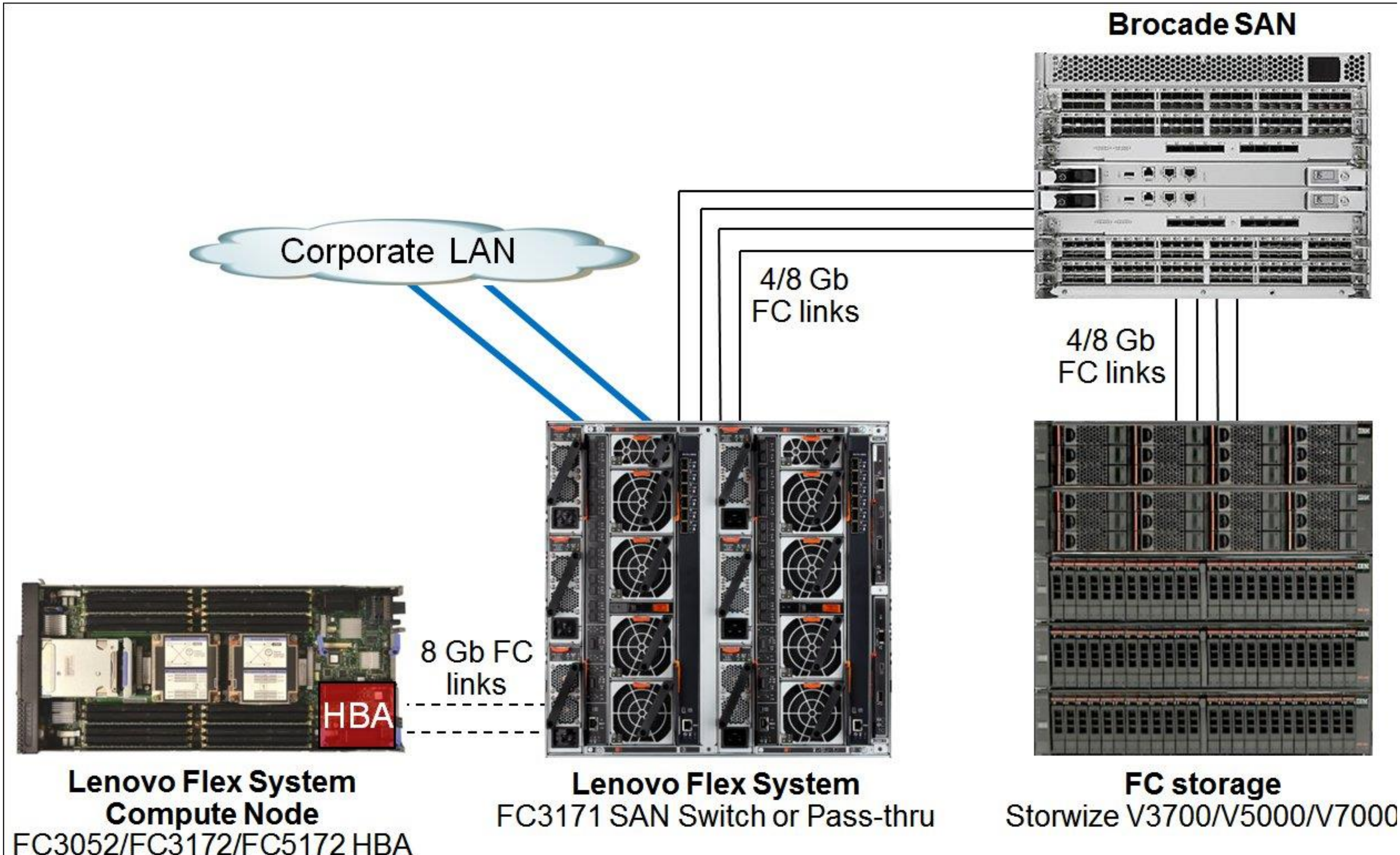
LAN for Data Center (SAN Storage)

Fibre Channel Switch + SAN



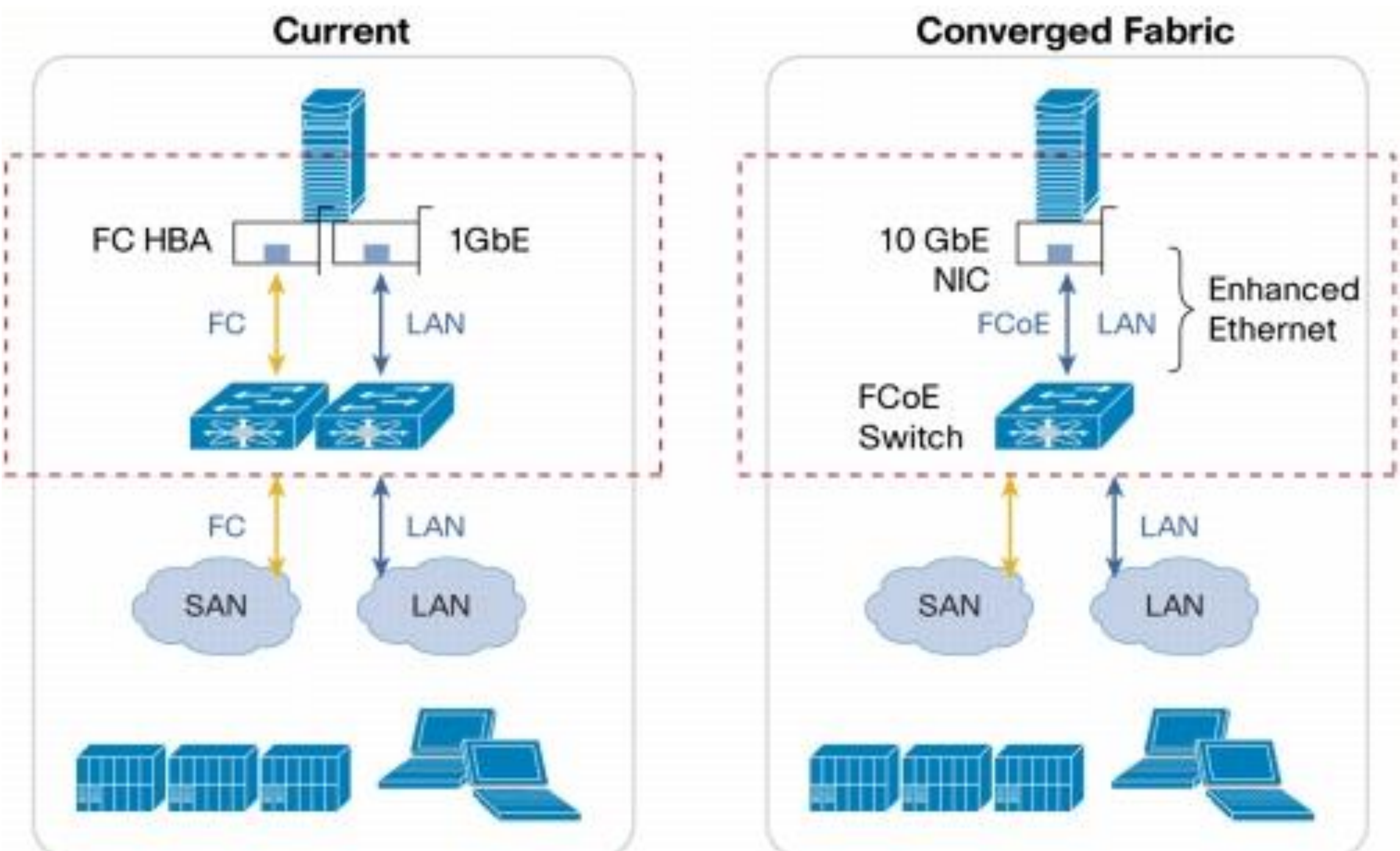
http://www.tsmtutorials.com/2016/08/fc-san-topologies.html

FC Switch + SAN Example



https://lenovopress.com/tips0866-fc3171-8gb-san-switch-and-pass-thru-modules

LAN for Data Center (FC vs FCoE)



http://e3tnw.org/ItemDetail.aspx?id=509